

Power BI Executive KPI Dashboard & Architecture Blueprint

Comprehensive Data Modeling, Power Query ETL, Star-Schema Design, and Executive DAX Measures

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Table of Contents

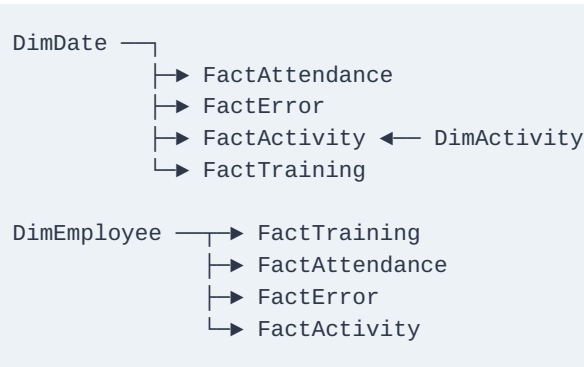
- 1. Recommended Power BI Data Architecture (Star-Schema Hubs)
- 2. Power Query Data Transformation & Cleaning (ID Alignment & Merges)
- 3. Unified Dimension Tables (DimEmployee, DimDate, DimActivity)
- 4. Model Relationships & Cross-Filtering (Cardinality & Direction)
- 5. Executive Report Page Layout (5-Page Dashboard Structure)
- 6. Core DAX Measures Catalog (Headcount, Training, Attendance, Errors, Activity & Overtime)

1. Recommended Power BI Data Architecture

To ensure robust query performance and prevent relational ambiguity, the raw transactional tables are structured into a **Star-Schema Executive Model** centered around `DimEmployee` and `DimActivity`.

Conceptual Model Topology

Rather than establishing direct connections across every transactional dataset, the architecture uses controlled dimension hubs:



2. Power Query Data Transformation & Cleaning

Before loading tables into the semantic model, standardizing keys and data types in Power Query is critical to prevent relationship failures.

Table Renaming Strategy

Source Table	Cleaned Name
Headcount Data	DimEmployee_Source
Staff Role Data	DimStaffRole_Source
Training Data 1 & 2	FactTrainingSession
Attendance Data	FactAttendance
Error Data	FactError
Activity Data	FactActivity
Activity TT Data	DimActivity

Key Normalization Rules

1. Staff IDs: Convert all employee identifiers (`Staff_No` , `Emp_ID` , `StaffID` , `Staff ID`) to whole numbers (Int64).

2. NTIDs: Ensure all NTID columns (`StaffNTID` , `ManagerNTID` , `ErrorCommitterNTID`) are formatted strictly as Text.

3. Supervisor ID Alignment: Standardize Headcount supervisor IDs to match Activity and Attendance formatting (e.g., prefixing integer `1023` with `M` to form `M1023`).

Power Query M Code for Manager ID Harmonization:

```
# "Added ManagerID" = Table.AddColumn("#Previous Step", "ManagerID", each "M" & Text.From([Job_Supervisor]), type text)
```

3. Unified Dimension Tables

To avoid orphan records (such as employees present in Activity logs but missing from static Headcount), a unified master employee dimension is established.

DAX Calculated Table: DimEmployee

```
DimEmployee =
DISTINCT (
    UNION (
        SELECTCOLUMNS ('Headcount Data', "StaffID", 'Headcount Data'[Staff_No],
            "StaffNTID", BLANK(), "EmployeeName", 'Headcount Data'[Employee_Name]),
        SELECTCOLUMNS ('Attendance Data', "StaffID", 'Attendance Data'[Staff ID],
            "StaffNTID", 'Attendance Data'[Staff NTID], "EmployeeName", 'Attendance Data'[Staff
            Name]),
        SELECTCOLUMNS ('Training Data 2', "StaffID", 'Training Data 2'[Emp ID],
            "StaffNTID", BLANK(), "EmployeeName", 'Training Data 2'[Emp_Name]),
        SELECTCOLUMNS ('Activity Data', "StaffID", 'Activity Data'[StaffID], "StaffNTID",
            'Activity Data'[StaffNTID], "EmployeeName", 'Activity Data'[StaffName])
    )
)
```

DAX Calculated Table: DimDate

```
DimDate =
VAR MinDate = MINX(UNION(SELECTCOLUMNS('FactAttendance', "D", 'FactAttendance'[Attendance
    Date]), SELECTCOLUMNS('FactError', "D", 'FactError'[ReportedDate]),
    SELECTCOLUMNS('FactActivity', "D", 'FactActivity'[ActivityDate]),
    SELECTCOLUMNS('FactTraining', "D", 'FactTraining'[Date Enrolled])), [D])
VAR MaxDate = MAXX(UNION(SELECTCOLUMNS('FactAttendance', "D", 'FactAttendance'[Attendance
    Date]), SELECTCOLUMNS('FactError', "D", 'FactError'[ReportedDate]),
    SELECTCOLUMNS('FactActivity', "D", 'FactActivity'[ActivityDate]),
    SELECTCOLUMNS('FactTraining', "D", 'FactTraining'[Date Enrolled])), [D])
RETURN
ADDCOLUMNS(
    CALENDAR(MinDate, MaxDate),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Year Month", FORMAT([Date], "YYYY-MM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Year Quarter", FORMAT([Date], "YYYY") & "-Q" & FORMAT([Date], "Q"),
    "Month Year", FORMAT([Date], "MMM YYYY")
)
```

4. Model Relationships & Cross-Filtering

All relationships must enforce **Single (1-to-Many)** cross-filter direction pointing from dimensions to facts, avoiding bidirectional performance hits and ambiguity.

From Table (1)	From Column	To Table (*)	To Column	Relationship Type
DimEmployee	StaffID	FactAttendance	Staff ID	1 to Many (Single)
DimEmployee	StaffID	FactTrainingSession	Emp ID	1 to Many (Single)
DimEmployee	StaffID	FactActivity	StaffID	1 to Many (Single)
DimActivity	ActivityID	FactActivity	ActivityID	1 to Many (Single)
DimDate	Date	FactAttendance	Attendance Date	1 to Many (Single)
DimDate	Date	FactError	ReportedDate	1 to Many (Single)
DimDate	Date	FactActivity	ActivityDate	1 to Many (Single)

5. Executive Report Page Layout

The Power BI report is structured into 5 distinct analytical pages designed for executive decision-making:

Page Name	Primary Focus	Key Visuals & Metrics
Page 1: Executive Overview	High-level organization health	Card KPIs (Headcount, Training %, Attendance %, OT Hours, Productivity %, Error Rate). Trend line of overall performance.
Page 2: Workforce & Organization	Demographics & headcount distribution	Bar charts by Unit, Grade, Manager, Location. Slicers for Employee Type, Nationality, and Gender.
Page 3: Training & Capability	Upskilling & compliance tracking	Completion % gauges, average training score matrices, high-priority training backlog, training hours by manager.
Page 4: Attendance & Errors	Operational risk & time tracking	Attendance % breakdown, late employee counts, OT breach flags, error points, and error type distribution.
Page 5: Productivity / Activity	Task execution & output efficiency	Activity volume vs. normal volume, productivity %, touch times, activity scores, and overtime score analysis.

6. Core DAX Measures Catalog

All measures should be centralized in a dedicated `_Measures` table for clean governance.

Headcount & Employee Measures

Total Employees = DISTINCTCOUNT (DimEmployee[StaffID])
Active Employees = CALCULATE ([Total Employees], DimEmployee[AssignmentStatus] = "Active")
Headcount = DISTINCTCOUNT (DimEmployee[StaffID])

```
Full Time Employees = CALCULATE ( [Total Employees], DimEmployee[EmployeeType] = "Full-Time" )
```

Training Measures

```
Training Records = COUNTROWS ( FactTrainingSession )
```

```
Completed Training = CALCULATE ( COUNTROWS ( FactTrainingSession ),  
FactTrainingSession[Training_Progress] = "Completed" )
```

```
Training Completion % = DIVIDE ( [Completed Training], [Training Records], 0 )
```

```
Average Training Score = AVERAGE ( FactTrainingSession[Assessment_Score] )
```

```
Training Hours = SUM ( FactTrainingSession[Training_Duration_(Hrs)] )
```

```
High Priority Training = CALCULATE ( COUNTROWS ( FactTraining ),  
FactTraining[Training_Priority] = "High" )
```

Attendance & Overtime Measures

```
Attendance Records = COUNTROWS ( FactAttendance )
```

```
Present Employees = CALCULATE ( DISTINCTCOUNT ( FactAttendance[Staff ID] ),  
FactAttendance[Attendance Type] = "Present" )
```

```
Attendance % = DIVIDE ( [Present Employees], [Attendance Records], 0 )
```

```
Total OT Hours = SUM ( FactAttendance[OT Hours] )
```

```
Paid OT Hours = SUM ( FactAttendance[Final Paid OT Hours] )
```

```
OT Breaches = CALCULATE ( COUNTROWS ( FactAttendance ), FactAttendance[Daily OT Breach Flag] =  
"Yes" )
```

```
Late Employees = CALCULATE ( DISTINCTCOUNT ( FactAttendance[Staff ID] ),  
FactAttendance[Exception_Type] = "Late" )
```

Error Risk Measures

```
Error Count = COUNTROWS ( FactError )
```

```
Error Points = SUM ( FactError[ErrorPoints] )
```

```
Average Error Points = AVERAGE ( FactError[ErrorPoints] )
```

```
Error Rate = DIVIDE ( [Error Count], [Total Employees], 0 )
```

```
Data Entry Errors = CALCULATE ( [Error Count], FactError[ErrorType] = "Data Entry Error" )
```

```
Compliance Errors = CALCULATE ( [Error Count], FactError[ErrorType] = "Compliance Error" )
```

Activity & Productivity Measures

```
Activity Records = COUNTROWS ( FactActivity )  
  
Employees with Activity = DISTINCTCOUNT ( FactActivity[StaffID] )  
  
Activity Score = AVERAGE ( FactActivity[Score] )  
  
Normal Score = AVERAGE ( FactActivity[NormalScore] )  
  
Activity Volume = SUM ( FactActivity[NoOfField] )  
  
Normal Volume = SUM ( FactActivity[NormalVolume] )  
  
Productivity % = DIVIDE ( [Activity Volume], [Normal Volume], 0 )  
  
Activity Performance % = DIVIDE ( [Activity Score], [Normal Score], 0 )  
  
Average Overtime Score = AVERAGE ( FactActivity[OvertimeScore] )
```